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Artificial Intelligence Seminar

*Design and implementation of a system
for the evaluation of the effectiveness of a
recommender system*

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1 Introduction

Barcelona is one of the clearest examples of what happens when tourism grows faster than a city can absorb it. The city received about 14.5 million visitors in 2024, and a small set of landmarks takes a large share of that pressure. The Sagrada Família alone reported 4,707,367 visitors in 2023, and Park Güell closed the same year with around 4.4 million [1], [2]. Tourist accommodation is spread just as unevenly, since two central districts, Ciutat Vella and the Eixample, together hold more than half of the city’s hotel and pension bed-places, while peripheral districts such as Nou Barris and Sant Andreu hold less than half a percent each [3]. The result is the familiar picture of overtourism, with crowded streets around a few icons, rising costs for residents, and whole districts that tourists rarely reach at all.

A recommender system can either make this worse or help to fix it. A system that always suggests the most popular places pushes even more people toward the same few spots. The seminar discussed a different idea, namely a multi-criteria recommender that is aware of sustainability. Such a system would still respect what the visitor wants, but it would also weigh social, cultural, economic and environmental factors, so that tourists are spread more evenly and lesser known places get a fair share of attention. As we saw in the points-of-interest seminar [4], this connects directly to concrete mechanisms such as rerouting tourists toward quieter attractions and tracking how they move through the city.

This work does not build that recommender; instead it asks a question that is easy to state and surprisingly hard to answer. If such a system already existed, how would we know that it actually improves the management of tourists, rather than simply assuming that it does? A recommender can look good on paper and still fail to change behaviour once real people react to crowds, distances and budgets. To study this we built an agent-based simulator. The simulator generates a large population of tourists, gives each of them a profile, lets several recommenders suggest places, and then simulates which places the tourists really visit. From those visits we measure crowding at each point of interest, the spread of tourists across neighbourhoods, and a set of recommender quality indicators. The central comparison is between a sustainability-aware recommender and three simpler baselines.

The contribution of this report is therefore an evaluation method rather than a new recommender. We describe a reproducible dataset of 77 real points of interest in Barcelona, four recommendation strategies, an agent-based model written with the Mesa framework, and a set of metrics grounded in the seminar material. We then report what the simulation shows, and the short version is that the sustainability-aware recommender does reduce overcrowding and does distribute visitors more evenly across districts, but it pays for this with a measurable drop in how well it matches each visitor’s stated interests. The rest of the report explains how we reached that conclusion and how far it can be trusted.

2 Background

2.1 Recommender systems and their evaluation

A recommender system filters a large set of items and proposes the ones that seem most relevant to a given user. As we saw in the introductory seminar [5], the field separates accuracy from what are often called beyond-accuracy properties. Accuracy metrics such as precision, recall and normalised discounted cumulative gain ask whether the system found the items the user would have chosen. Beyond-accuracy metrics look at the things that still matter once those obvious matches are covered. Diversity captures how different the recommended items are from one another, novelty rewards a system that is willing to suggest less popular items, and coverage tracks how much of the full catalogue ever gets recommended at all. These properties matter here because a sustainability goal is, in large part, a beyond-accuracy goal, given that spreading tourists out is the opposite of always recommending the same famous landmarks.

2.2 Tourism recommenders and overtourism

Tourism is a demanding setting for recommenders because the items are physical places with limited capacity, opening hours and a location in space. The points-of-interest seminar framed the sustainability problem in practical terms [4]. It described three levers that a tourism recommender can pull, namely rerouting visitors toward less crowded attractions, tracking how people move so that the city can react, and regulating which attractions are shown to which visitors. All three assume that the recommendations actually change where people go, and that assumption is exactly what a simulation is able to test.

2.3 Multi-criteria decision aiding

The sustainability-aware recommender rests on multi-criteria decision aiding. As we saw in the multi-criteria seminar [6], a recommendation can be treated as a ranking problem over several criteria that are not directly comparable, for example a visitor’s interest, the crowding of a place, and its environmental quality. The seminar presented the ELECTRE family of outranking methods, which compare alternatives pair by pair using concordance and discordance, together with simpler weighted aggregation. The companion sustainable-tourism material organised the criteria into four pillars that cover environmental, cultural, social and economic concerns. We follow that structure when we build the criteria for our recommender, and we use ELECTRE-III as the ranking method because it was the approach presented in that seminar.

2.4 Context, fairness and trust

The way we evaluate the system is shaped by three more seminars. The context-aware seminar [7] described how factors such as the group, the budget and the available

time change what counts as a good recommendation, which is what motivates the rich tourist profiles we generate. The trustworthy systems seminar [8] listed properties such as transparency, explainability and fairness, and it made a point that is central to our design, since offline evaluation misses the feedback loop that appears once real recommendations change behaviour, which in turn changes the future data. An agent-based simulation is one way to put that loop back in. Finally, the knowledge-graph seminar [9] treated fairness as a question of exposure, where some items are shown far more often than others. We measure that directly with a Gini coefficient over how often each place is recommended.

3 Data

3.1 Sources and integrity

We built a dataset of 77 points of interest in Barcelona. Every populated field traces back to a citable source, and any field without a reliable source was left empty rather than guessed. Names, coordinates and heritage status were taken from Wikidata [10], while the exact Wikipedia article titles and their monthly page views were pulled from the Wikimedia Pageviews API [11] and summed over the twelve months from June 2025 to May 2026. For accessibility and fee tags we queried OpenStreetMap through the Overpass API [12]. District and neighbourhood labels follow the official municipal cartography through a point-in-polygon assignment, the district areas are read from the 2024 statistical yearbook [13], and the visitor counts for a handful of major attractions come from each site’s own reports and from municipal press releases [1], [2], [14].

3.2 Schema and derived criteria

Each point of interest carries a category from a fixed list, its coordinates, its district and neighbourhood, a popularity signal, an entrance fee, a green-space flag, an accessibility label and a heritage-protection level. From these raw fields we compute the criteria that the recommenders use. Popularity is the page-view total rescaled to the range zero to one, the environmental score is simply the green-space flag, and accessibility maps the OpenStreetMap labels to a numeric score, while cultural value maps the Catalan heritage levels (national, local, or none) onto a score of its own.

Two fields deserve a clear explanation, because the data forced design choices. First, no point of interest publishes an official daily visitor cap, so the capacity field is empty for all of them. We therefore use a single flat capacity for every place, kept deliberately independent of popularity so that crowding emerges from demand and not from the rule itself. Second, no official figure for tourists per square kilometre is published by district. We replace it with a measure we can actually source, namely the district’s share of the city’s tourist bed-places divided by its area [3], [13]. This ranks Ciutat Vella above the Eixample, which matches the known pattern of pressure in the city. From this we define

a low-pressure score, higher for calmer districts, and a local-economy score based on each place’s distance from the central tourist core around Plaça de Catalunya. Both choices are simplifications, and we return to them in the limitations.

4 System design

4.1 Tourist profiles

A profile describes a single tourist, and it holds a weight for each category of interest, a budget for entrance fees, a mobility mode (walking or public transport), a tolerance for walking, a crowd aversion, a sustainability sensitivity, a preference for outdoor places, whether the group travels with children, the group size, the planned number of visits, and the district where the trip starts. The population generator samples these values with a fixed seed, so the same population can be replayed under every recommender. Most tourists care about one to three categories more than the rest, which produces the kind of varied demand that a real city sees.

4.2 The four recommenders

All four recommenders share the same interface, in that each one receives a tourist profile together with the set of places and returns a ranked list of place identifiers.

The popularity recommender ranks places by their page-view popularity and ignores the profile completely. It is the strategy a naive system would use, and it is the one most likely to send everyone to the same icons.

The interests recommender ranks places by how well their category matches the profile, after removing places whose fee exceeds the budget. It is a strong personalised baseline that should score well on accuracy.

The sustainability-aware recommender is the system under test. It scores each place on six criteria, namely the fit with the visitor’s interests, the low-pressure score of its district, its environmental score, its accessibility, its cultural value, and its local-economy score. The weight given to preference against sustainability depends on the visitor. If s is the sustainability sensitivity of the profile, the preference criterion receives weight $1 - 0.6s$, and the five sustainability criteria share the remaining weight equally, so each receives $0.6s/5$. A visitor with no sustainability sensitivity is ranked almost purely on interest, while a very sensitive visitor still keeps a meaningful share of weight on preference. The places are then ranked with ELECTRE-III, using the implementation in the pyDecision library [15], after a cheap weighted-average pre-selection that keeps the method fast on the full catalogue. A weighted-average ranking is kept as a cross-check, which also matches the simpler aggregation shown in the multi-criteria seminar [6].

The random recommender returns the places in a random order, and it serves as a lower bound that helps to make sense of every other number in the comparison.

4.3 The agent-based model

The simulation is built on the Mesa framework [16], and each tourist is modelled as a single agent. At the start of the trip the agent asks the active recommender for its ranked list once, which mirrors a real visitor who receives a set of suggestions and then works through them over the day. At every step the agent looks at the next few unvisited and affordable suggestions and picks one of them, and that choice is not purely greedy. It trades off the rank of the place, its current crowding, and the travel cost from the agent’s current location, with crowding weighted by the agent’s own crowd aversion and travel weighted against its walking tolerance and mobility mode. The agent then visits the chosen place, which updates its budget, its location, and the crowding counters of the place and its district.

The city model holds the shared state of the simulation, with a live count of visits at every place and in every district and a record of each place’s capacity. Once a place passes its capacity it becomes less attractive, so the later crowd-averse tourists tend to go elsewhere instead. This is the feedback loop that the trustworthy systems seminar argued is missing from static evaluation [8]. Every strategy is run on the same population with the same seed, so the only thing that changes between runs is the recommender.

4.4 Interactive demo

The repository also includes an interactive dashboard built with Streamlit, which turns the results of the simulation into something a reader can explore directly. The dashboard draws the points of interest on a map of Barcelona, where the size and colour of each marker grow with how crowded that place becomes under the chosen recommender. Switching the recommender updates the map, so the contrast between the popularity strategy and the sustainability-aware one is easy to see. The same view shows the metric comparisons from this report as charts, together with the split of visits across the ten districts. The dashboard runs locally with the command `streamlit run demo/streamlit_app.py`, and it can be published as a web application in the same way as any other Streamlit project. Figures 1 and 2 show the dashboard while it is being used.

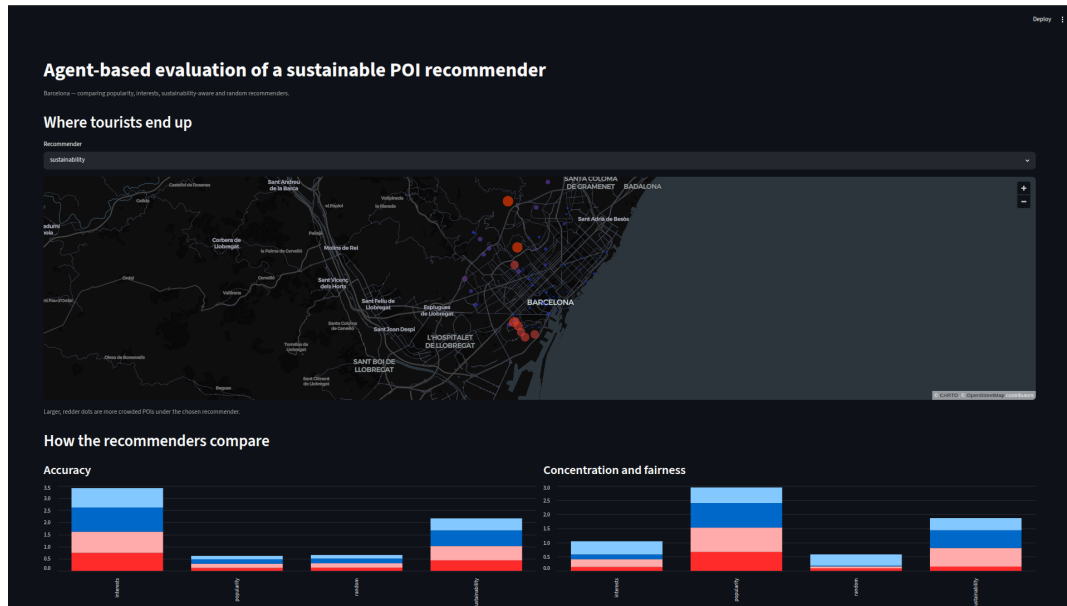


Figure 1: The Streamlit dashboard, with the crowding map of Barcelona for the selected recommender and the first of the metric comparisons starting below it.



Figure 2: The lower part of the dashboard, with the diversity, coverage and sustainability charts, the spread of visits across the ten districts, and the full metrics table.

5 Experimental setup

The simulation runs with a population of 5,000 tourists, and each tourist plans between three and eight visits, so a single run produces on the order of twenty-seven thousand visits spread over the 77 places. We run the four recommenders on this single population and record, for each one, the full recommendation list of every tourist, the places each tourist actually visited, and the final crowding state of the city.

The metrics fall into four groups, and we describe each of them in turn. The first group, accuracy, uses precision, recall, the F1 score and normalised discounted cumulative gain at a cut-off of ten. A place counts as relevant to a tourist when it belongs to a category that the tourist clearly likes, which follows the category-based relevance used in the points-of-interest seminar [4]. Beyond-accuracy uses intra-list diversity, novelty and catalogue coverage. Intra-list diversity is the average dissimilarity between pairs of recommended places, combining a category difference and a geographic distance. Novelty is the average of $-\log_2 p_i$ over the recommended places, where p_i is the normalised popularity of place i , so rarely visited places score higher. Coverage is the fraction of the catalogue that appears in any recommendation.

The third group measures fairness and concentration, for which we use the Gini coefficient of the visit counts across places, the Gini of visits across districts, and the Gini of how often each place is recommended, which is the exposure fairness from the knowledge-graph seminar [9]. We also count how many places exceed their capacity, and we report the share of all visits that land on the most popular tenth of the catalogue. The fourth group measures the sustainability of the visits that actually happened. For each sustainability criterion we compute its average over visited places, weighted by the number of visits, and we report the share of visits that reach low-pressure districts.

6 Results

Table 1 reports the accuracy and beyond-accuracy metrics, and Table 2 reports the fairness, concentration and sustainability metrics. All numbers come from the same 5,000-tourist run.

Table 1: Accuracy and beyond-accuracy metrics by recommender.

Metric	Popularity	Interests	Sustainability	Random
Precision@10	0.178	0.878	0.591	0.181
Recall@10	0.117	0.738	0.431	0.131
F1@10	0.141	0.802	0.499	0.152
NDCG@10	0.181	0.998	0.645	0.191
Diversity (ILD)	0.535	0.333	0.526	0.606
Novelty	4.58	9.05	8.62	9.56
Coverage	0.130	0.961	0.974	1.000
Personalisation	0.000	0.861	0.665	0.872

Table 2: Fairness, concentration and sustainability outcomes by recommender. Lower is better for the Gini rows, the overcrowding count and the top-popular share. Higher is better for the visited-criteria rows.

Metric	Popularity	Interests	Sustainability	Random
POI visit Gini	0.862	0.277	0.658	0.070
District visit Gini	0.552	0.472	0.432	0.404
Exposure Gini	0.870	0.162	0.626	0.020
Overcrowded POIs	9	0	5	0
Top-popular visit share	0.667	0.135	0.151	0.088
Visited low-pressure	0.535	0.517	0.860	0.558
Visited environmental	0.081	0.119	0.254	0.136
Visited accessibility	0.798	0.689	0.754	0.661
Visited cultural	0.815	0.583	0.728	0.561
Visited local-economy	0.275	0.308	0.473	0.334
Low-pressure district share	0.235	0.317	0.641	0.336

6.1 Accuracy

The accuracy results are the least surprising part of the study, and they are shown in Figure 3. The interests recommender wins on every accuracy metric, with a precision at ten of 0.88 and a near-perfect ranking quality of 1.00. This is expected, since that recommender optimises directly for category match. The sustainability-aware recommender sits in the middle with a precision of 0.59 and a ranking quality of 0.65. The popularity and random strategies fall far below, both near 0.18, because matching the average famous place to one specific visitor’s interests is mostly a matter of luck. The reading here is fairly clear, since adding sustainability criteria does cost accuracy, yet the sustainability-aware recommender still keeps a clear and useful link to what each

visitor wants and stays well above the popularity baseline that many real systems rely on.

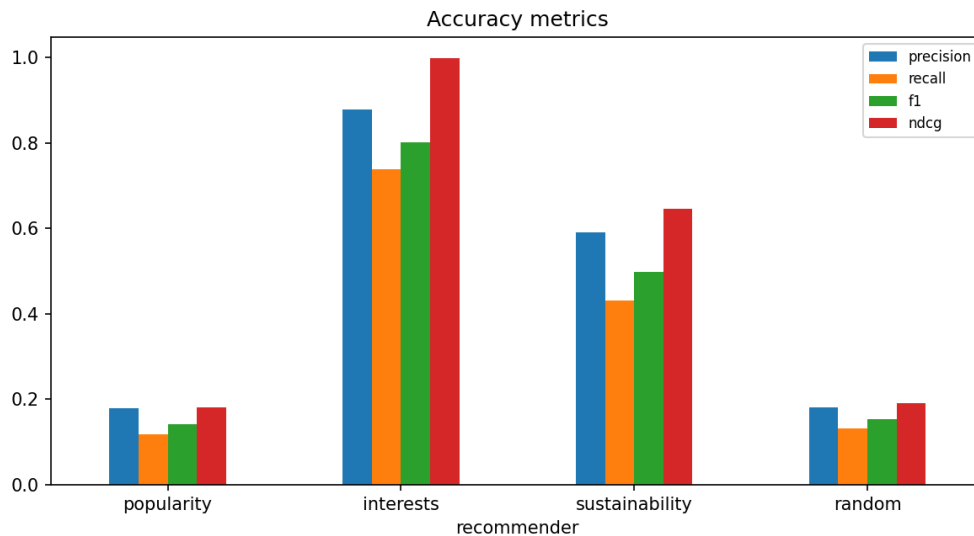


Figure 3: Accuracy metrics by recommender. The interests recommender leads, the sustainability-aware recommender follows, and popularity and random trail.

6.2 Concentration and overcrowding

Concentration is where the sustainability argument starts to pay off, as Figure 4 shows. Under the popularity recommender, two thirds of all visits land on the most popular tenth of the places, the visit Gini reaches 0.86, and nine places exceed their capacity. The sustainability-aware recommender changes this picture, since the top-popular share falls from 0.67 to 0.15 and the number of overcrowded places drops from nine to five. The interests recommender also lowers concentration, simply because varied interests scatter people across many places. The point worth stressing is that the popularity strategy, which is the easy default, is by far the worst for crowding, and the sustainability-aware strategy is the one designed to fix it.

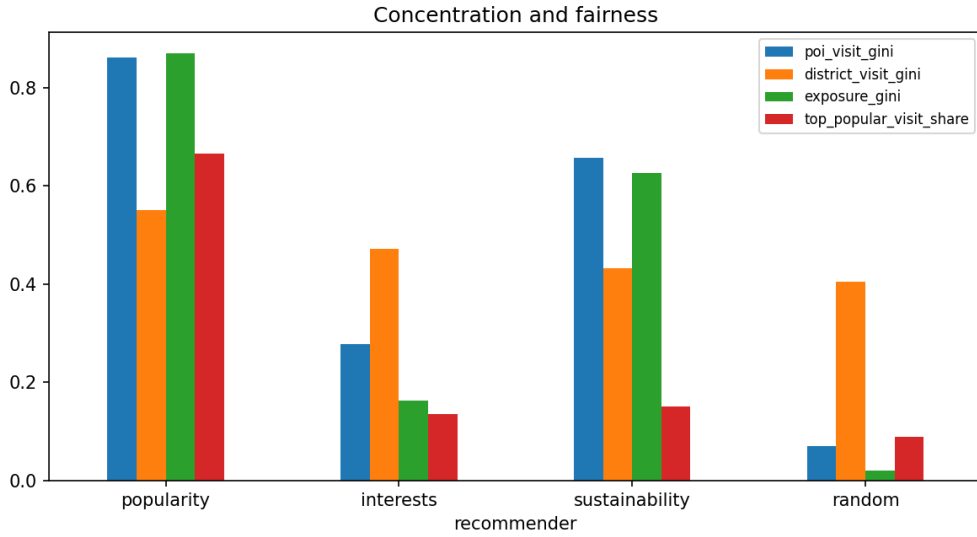


Figure 4: Concentration and fairness by recommender. Popularity concentrates both visits and exposure; the sustainability-aware recommender reduces both, though not as far as the interests baseline.

There is an honest tension in these numbers that is worth spelling out. The sustainability-aware recommender has a higher place-level visit Gini (0.66) than the interests recommender (0.28), which looks contradictory at first but is not, once we read it carefully. The recommender deliberately funnels visitors toward a curated subset of sustainable places, so visits do pile up on those particular places, even though the spread across the city as a whole still improves. The district results, which we turn to next, confirm exactly that wider geographic spread.

6.3 Distribution across the city

The clearest sustainability win is spatial, and it appears in both the district metrics and the maps. The sustainability-aware recommender sends 64 percent of visits to low-pressure districts, against only 23 percent for the popularity recommender and 32 percent for the interests recommender. Its district visit Gini, at 0.43, is lower than both popularity (0.55) and interests (0.47). Figure 5 shows how each strategy splits visits across the ten districts, and Figure 6 shows the same effect on the map. Under popularity, a handful of central places swell while the rest of the city stays empty. Under the sustainability-aware recommender, the large markers move outward, toward Montjuïc, the upper districts and the periphery. This is the behaviour the city would want from a system meant to relieve the centre.

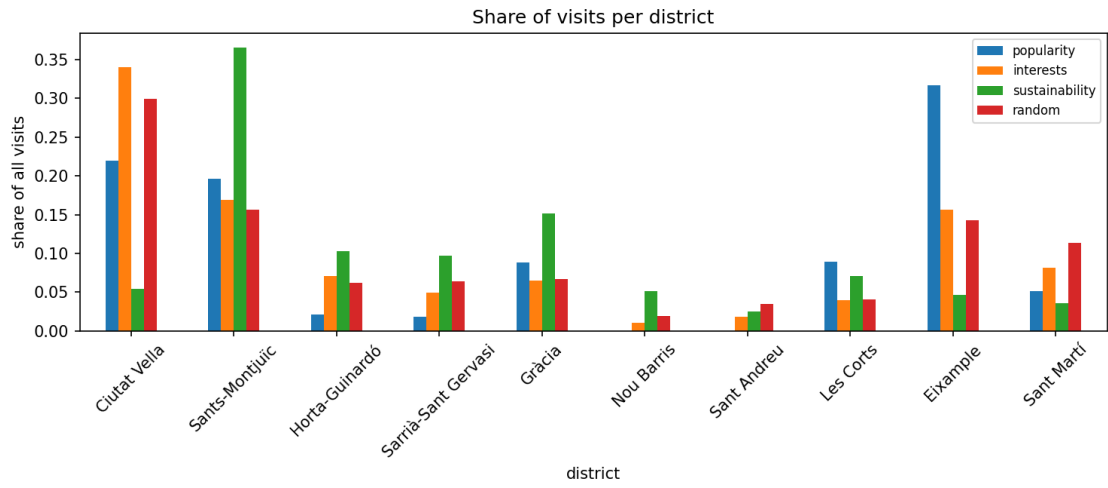


Figure 5: Share of visits per district. The popularity recommender concentrates visits in a few central districts, while the sustainability-aware recommender pushes a larger share toward calmer districts.

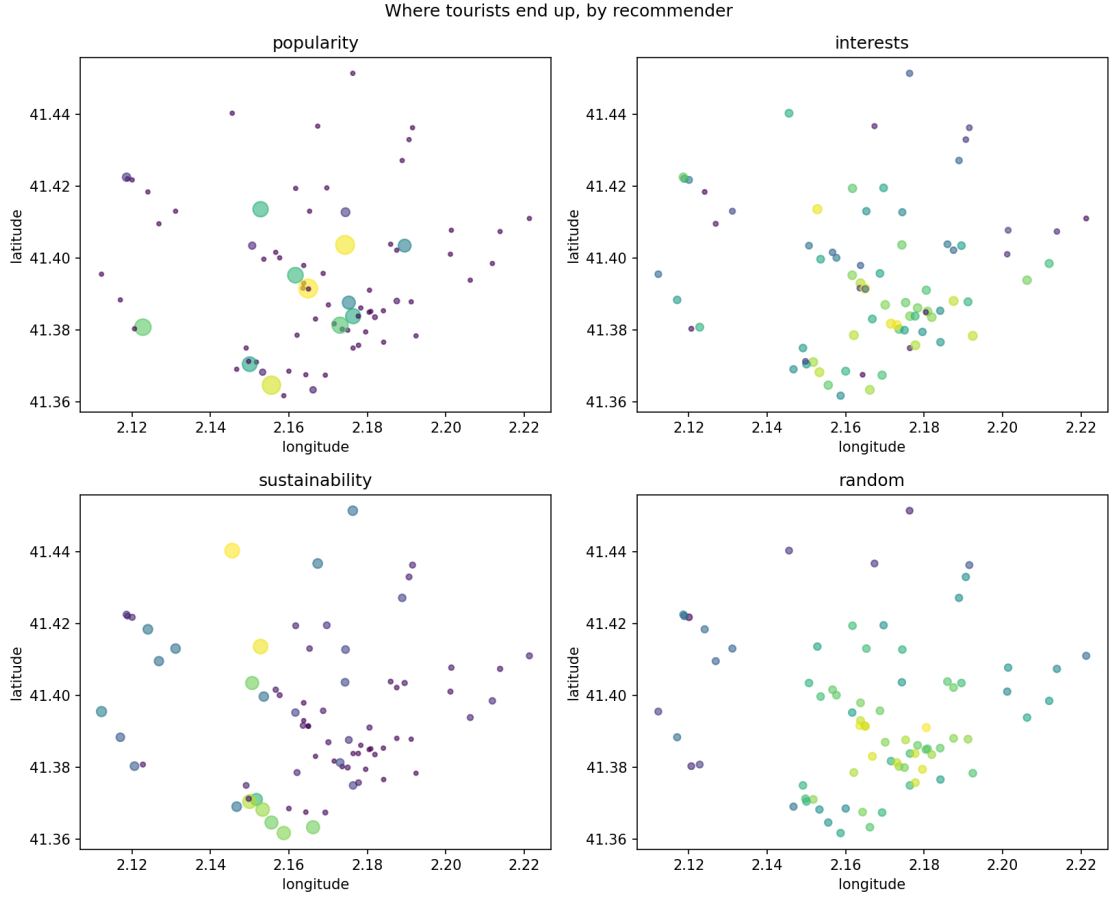


Figure 6: Where tourists end up under each recommender. Marker size and colour grow with the number of visits. Popularity piles visitors on a few central giants; the sustainability-aware recommender spreads larger markers toward the edges of the city.

6.4 Sustainability of the visited places

The last group of metrics looks at the quality of the places people actually reached, and it is summarised in Figure 7. The sustainability-aware recommender leads on the sustainability criteria it was built to serve. Visits under this strategy reach places with the highest average low-pressure score (0.86 against 0.53 for popularity), the highest environmental score (0.25 against 0.08), and the highest local-economy score (0.47 against 0.28). It also keeps cultural and accessibility scores high. These gains are consistent with the design, since these are the criteria the recommender optimises, and they show that the optimisation survives the choices that tourists make once crowding and distance enter the picture.

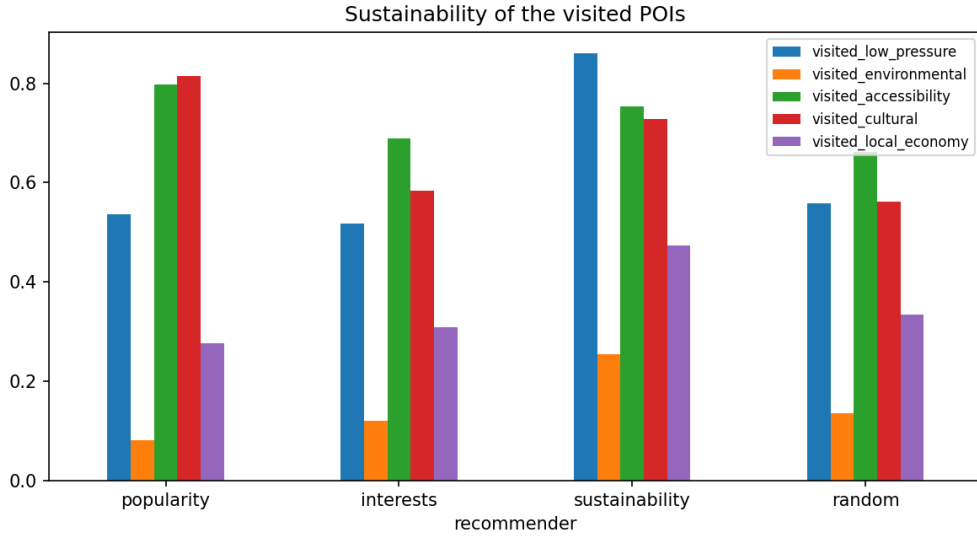


Figure 7: Average sustainability scores of the visited places, weighted by visits. The sustainability-aware recommender leads on the criteria it optimises.

Novelty and coverage round out the picture in the same direction. The popularity recommender has the lowest novelty (4.58) and a coverage of only 0.13, which means it leans on a small fixed set of famous places, whereas the sustainability-aware recommender reaches a coverage of 0.97 and a high novelty of 8.62, close to the random baseline, so it genuinely surfaces places that visitors would not otherwise see.

7 Further analyses grounded in the seminars

Beyond the main four-way comparison, we ran three smaller studies, each tied to a specific seminar. They keep the main comparison clean while showing that the system can be pushed in the directions the seminars pointed to.

7.1 Context-aware recommendation

The context-aware seminar [7] separated pre-filtering from post-filtering, and we take the post-filtering route here. We gave every tourist a weather context and added a step that pushes outdoor places to the back of the list on rainy days. Table 3 shows the effect on the share of outdoor visits made by rainy-weather tourists. Without the context step, rainy tourists still send about 29 percent of their visits to outdoor places such as parks, beaches and viewpoints. With it, that share falls to zero, because the outdoor places no longer reach the top of the list. The post-filter adapts the recommendation to the weather without changing the underlying ranking model, which is the point the seminar made about post-filtering.

Table 3: Share of visits to outdoor POIs among rainy-weather tourists.

Recommender	Rainy-day outdoor visit share
Sustainability	0.289
Sustainability + context	0.000

7.2 Tourist segmentation

The tourism seminar [4] grouped tourists into segments before studying their behaviour. We clustered the 5,000 tourists with k-means into five segments and named each one by the category its members care about most. Table 4 and Figure 8 report, for each segment, its size together with two outcomes under the sustainability recommender, namely the precision of the recommendations and the average low-pressure score of the visited places. The recommender does not serve every segment equally. Tourists who like architecture or parks receive the most accurate recommendations, near 0.79 to 0.81, while the large museum segment receives the least accurate, around 0.46, because museums compete with many sustainable alternatives. The low-pressure score stays high across all segments, between 0.77 and 0.93, so each segment is still steered toward calmer districts. We also added the personalisation metric from the same work, shown earlier in Table 1. It is zero for the popularity recommender, which hands everyone the same list, and high for the other three, which tailor the list to each tourist.

Table 4: Per-segment outcomes under the sustainability recommender.

Segment (dominant interest)	Size	Precision	Visited low-pressure
museum	2044	0.456	0.874
market	768	0.552	0.847
square landmark	736	0.595	0.774
park / nature	736	0.786	0.929
architecture	716	0.814	0.888

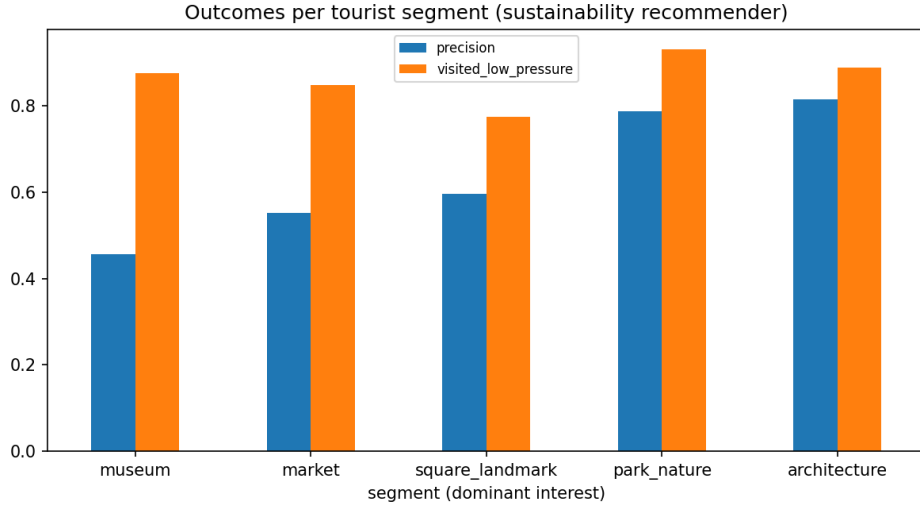


Figure 8: Precision and visited low-pressure score per tourist segment, under the sustainability recommender.

7.3 Choice of aggregation method, and explanations

The multi-criteria seminar [6] showed both ELECTRE and simpler weighted aggregation. We ran the sustainability recommender with each method on the same population and compared them in Table 5. The two methods agree on about 61 percent of their top ten places, but they differ where they disagree. The weighted average reaches a much higher precision (0.88 against 0.59) and a slightly lower concentration, because it respects the large weight that the profile puts on the visitor’s interest. ELECTRE-III instead pushes harder toward low-pressure districts (0.64 against 0.49), and the reason is structural rather than accidental. ELECTRE compares places criterion by criterion, so the five sustainability criteria can together outrank a single preference criterion even when that one criterion carries the largest individual weight. This matters in practice, because it shows that the popular outranking method is not automatically the better choice, and that a simpler weighted average is a serious alternative for this task.

Table 5: ELECTRE-III against a weighted average, same criteria and weights.

Metric	ELECTRE-III	Weighted average
Precision@10	0.591	0.878
Top-popular visit share	0.151	0.102
District visit Gini	0.432	0.270
Low-pressure district share	0.641	0.489
Top-10 overlap	0.609	

The trustworthy systems seminar [8] asked for recommendations that can be explained, and the knowledge-graph seminar [9] made the same point through criterion-level reasoning. Because our recommender scores each place on named criteria, every suggestion can be shown as a set of criterion values. Three examples in Table 6 make this concrete. The museum lover receives the National Art Museum, which fits the interest and also scores well on most sustainability criteria, so interest and sustainability agree. The religious-site lover instead receives Parc del Laberint d’Horta, a quiet historic garden in a low-pressure district. Here the interest fit is low, only 0.18, but the place scores at the top on low pressure, environment and local economy, so the recommender trades interest for sustainability. This is the redistribution the system is meant to produce, and the criterion values make the reason visible to the user.

Table 6: Criterion values behind the top recommendation for three example tourists.

Tourist	Top recommen- dation		Pref.	Low-pr.	Env.	Acc.	Cult.	Local
museum lover	National Art Museum (mu- seum)		0.86	0.94	0.00	1.00	1.00	0.35
religious-site lover	Parc del Laberint d’Horta (park)		0.18	0.99	1.00	0.50	0.70	0.87
nature lover	Parc del Laberint d’Horta (park)		0.86	0.99	1.00	0.50	0.70	0.87

8 Discussion

8.1 What the simulation shows

The results support a careful version of the seminar’s claim. A sustainability-aware recommender can reduce overcrowding and can distribute visitors more evenly across a city, and it does so while keeping recommendation quality far above a popularity baseline. The trade-off is real and should not be hidden. The interests recommender remains the most accurate, and the sustainability-aware recommender gives up part of that accuracy in exchange for better distribution and better sustainability scores. A city that cares about overtourism would likely accept that exchange, while a purely commercial product might not. The decision is a policy choice, and the value of the simulator is that it puts numbers on both sides of it.

9 Conclusion

We set out to test whether a sustainability-aware recommender would actually improve the management of tourists in a city, rather than only look good in theory. Using an agent-based simulation of 5,000 tourists over 77 real points of interest in Barcelona, we compared such a recommender against popularity, interests and random baselines. The sustainability-aware recommender cut the share of visits to the most popular places from two thirds to about a sixth, reduced the number of overcrowded places, and raised the share of visits reaching calmer districts from under a quarter to nearly two thirds. It did this while keeping recommendation quality clearly above the popularity baseline, though below the purely interest-driven one. Within the limits of a simulated population, the evidence supports the seminar’s idea, since weighing sustainability alongside preference moves tourists away from the saturated centre and toward the rest of the city, at a cost in accuracy that a city focused on overtourism would probably be willing to pay.

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